Rubric: Triangle and Season Classification Lab (30 Points)

1. Boolean Expression Evaluation (5 Points)

Objective: Evaluate simple and nested boolean expressions.

Criteria	Points	Description
Correct evaluation of simple expressions	5	Accurately evaluates all boolean expressions (e.g., `(A + B) > C`).

2. Triangle Classification Program (10 Points)

Objective: Implement a program to classify triangles and check for Pythagorean triples.

Criteria	Points	Description
Input collection and validation	2	Prompts for and validates user input (e.g., positive numbers, non-numeric input).
Triangle validity check	2	Correctly checks if sides form a valid triangle using the triangle inequality theorem.
Triangle classification	3	Accurately classifies triangles as equilateral, isosceles, or scalene.
Pythagorean triple check	2	Correctly identifies if sides form a Pythagorean triple.
Output formatting and messages	1	Prints clear and correct messages for all cases (valid/invalid triangle, Pythagorean triple).

3. Date Validation and Season Determination Program (10 Points)

Objective: Implement a program to validate dates and determine seasons.

Criteria	Points	Description
Input collection and validation	2	Prompts for and validates user input (e.g., correct date format, valid month/day).
Date validation	3	Correctly checks if the month and day are valid (e.g., `month in 1–12`, valid days for each month).
Season determination	4	Accurately determines the season based on the date (e.g., Spring, Summer, Autumn, Winter).
Output formatting and messages	1	Prints clear and correct messages for all cases (valid/invalid date, season).

4. Code Quality and Style (5 Points)

Objective: Practice good coding style and organization.

Criteria	Points	Description
Header comment block	1	Includes a header comment block with a description of the problem.
Code readability and organization	2	Uses appropriate white space, indentation, logical organization and variable names
Comments and documentation	2	Includes meaningful comments to explain key logic and steps.

Grading Scale

- **27–30 Points:** Excellent – All objectives met with exceptional quality and accuracy.
- **24–26 Points:** Good – Most objectives met with minor issues or omissions.
- **18–23 Points:** Satisfactory – Some objectives met, but significant issues or omissions exist.
- **0–17 Points:** Needs Improvement – Major issues or incomplete work.